Email Forensic Analysis Report

Case EFT-20261028-312DF59E - generated 2026-10-28T06:55:57+00:00

Case identifier	EFT-20261028-312DF59E
Analysis time (UTC)	2026-10-28T06:55:57.841487+00:00
Report generated (UTC)	2026-10-28T06:55:57+00:00
Tool	Email Forensics Toolkit v1.0.0
Source	L09_forwarded_spf_fail.eml
Input method	file upload
Evidence size	1125 bytes
SHA-256 (at intake)	807bb5d2c582abfd10e2919af44ab1671a4a8116a04c13ffd433705e21742b9f
Integrity re-check	INTACT - stored original matches the intake hash (verified 2026-10-28T06:55:57+00:00)
Analysis settings	scoring=severity_sum; trusted domains=['uni-example.edu']; trusted authserv-ids=auto; extra protected domains=none; DNS lookups=off; clock-skew tolerance=5 min

1. Risk assessment

Category: **LOW** Score: **1** Approach: severity_sum (thresholds medium ≥ 3, high ≥ 6)

The score ranks how many and how serious the header indicators are. It is not a calibrated probability of maliciousness, and a low score does not prove a message is safe (e.g. a compromised legitimate account passes every header check).

Rule	Finding	Severity	Points	Why these points
AUTH-SPF-FAIL	SPF fail for the envelope sender	low	1	low severity = 1 point(s)

Confidence note: Message was relayed (list/forwarder); some failures are expected and were down-weighted.

2. Extracted header information

From	"Omar Farooq" <omar.farooq@outlook.com>
Sender	-
Reply-To	-
Return-Path	<omar.farooq@outlook.com>
To	<student@uni-example.edu>
Cc	-
Subject	Reunion photos
Date	Mon, 14 Sep 2026 09:28:00 +0000 (parsed: 2026-09-14T09:28:00+00:00)
Message-ID	<811965672782.299624@outlook.com>
Domains observed	outlook.com
IP addresses	10.20.0.11 (private), 64.62.200.3 (public), 40.107.22.98 (public)
Header fields	13 total; 3 Received; full message

3. Apparent delivery route (earliest first)

Trusted infrastructure: uni-example.edu. Origin IP seen by trusted server: 64.62.200.3 (public). Hops marked 'claimed' were written by servers outside the trust boundary and may be forged.

#	From (HELO / rDNS / IP)	By	Timestamp	Delay	Trust
1	eur05-am6-obe.outbound.protection.outlook.com / eur05-am6-obe.outbound.protection.outlook.com / 40.107.22.98	mx.alumni-forward.org	2026-09-14T09:29:00+00:00	-	claimed
2	relay.alumni-forward.org / relay.alumni-forward.org / 64.62.200.3	mx1.uni-example.edu	2026-09-14T09:30:00+00:00	+60 s	trusted
3	mx1.uni-example.edu / mx1.uni-example.edu / 10.20.0.11	mailbox01.uni-example.edu	2026-09-14T09:30:12+00:00	+12 s	trusted

4. Authentication results

Mechanism	Status	Reported	Domain	Interpretation
SPF	**fail**	fail	outlook.com	The connecting IP is NOT authorised by the envelope sender's SPF record. Common for spoofing, but also for forwarded mail.
DKIM	**pass**	pass	outlook.com	A DKIM signature verified, so the signed headers/body were not altered after signing by the signing domain (d=).
DMARC	**pass**	pass	outlook.com	SPF or DKIM passed for a domain aligned with the visible From domain.

Alignment (computed): From=outlook.com, SPF domain=outlook.com (aligned: False), DKIM pass domains=outlook.com (aligned: True). Relaxed alignment computed by this tool from the trusted results; it approximates, but is not, a DMARC evaluation.

Trusted authserv-ids: mx1.uni-example.edu; untrusted results ignored: 0.

5. Findings

1. SPF fail for the envelope sender [LOW] AUTH-SPF-FAIL

Evidence: spf=fail smtp.mailfrom domain=outlook.com

Why it matters: The server that delivered the message is not authorised to send for the envelope-sender domain, which is consistent with spoofing.

Limitations: SPF checks the Return-Path domain, not the visible From. Forwarding breaks SPF for legitimate mail.

Mitigating context: forwarding headers: X-Forwarded-To

2. Message shows forwarding / ARC markers [INFO] CONTEXT-FORWARDED

Evidence: forwarding headers: X-Forwarded-To

Why it matters: Forwarders send from IPs not listed in the original sender's SPF record, so SPF failure is expected; ARC can carry the original results.

Limitations: ARC is only as trustworthy as the intermediaries that sealed it.

6. Limitations

- Headers can show that evidence supports or weakens a hypothesis; they cannot prove who wrote a message or whether an account was compromised.

- Received headers below the trust boundary are supplied by the sending side and may be forged.

- Authentication results are interpreted as recorded by the receiving server; signatures are not re-verified by this tool.

- Organisational domains use a built-in suffix list, not the full Public Suffix List; look-alike detection is heuristic and limited to protected domains.

- Message bodies, links and attachments are not analysed.

- Auto-trusted authserv-id 'mx1.uni-example.edu' (topmost Authentication-Results, matching the receiving server's domain).

7. Recommended next steps

- Check the From domain's current SPF/DMARC records and compare with the receiving server's logs for this Message-ID.

- Correlate the origin IP and Message-ID with mail-server and gateway logs.

- Preserve this report together with the original file; verify the SHA-256 hash before any further processing.

Appendix A - Header fields as parsed (unfolded, in original order)

Control characters are shown as U+FFFD and very long values are truncated. The unmodified original is preserved separately under the SHA-256 above.

```
Received: from mx1.uni-example.edu (mx1.uni-example.edu [10.20.0.11]) by mailbox01.uni-example.edu with LMTP id
    8F2A1C; Mon, 14 Sep 2026 09:30:12 +0000
Authentication-Results: mx1.uni-example.edu; spf=fail smtp.mailfrom=outlook.com; dkim=pass header.d=outlook.com;
    dmarc=pass header.from=outlook.com
Received: from relay.alumni-forward.org (relay.alumni-forward.org [64.62.200.3]) by mx1.uni-example.edu with ESMTPS id
    06617277 for <student@uni-example.edu>; Mon, 14 Sep 2026 09:30:00 +0000
Received: from EUR05-AM6-obe.outbound.protection.outlook.com (EUR05-AM6-obe.outbound.protection.outlook.com
    [40.107.22.98]) by mx.alumni-forward.org with ESMTPS id 17153437 for <student@uni-example.edu>; Mon, 14 Sep 2026
    09:29:00 +0000
Return-Path: <omar.farooq@outlook.com>
From: "Omar Farooq" <omar.farooq@outlook.com>
To: <student@uni-example.edu>
Subject: Reunion photos
Date: Mon, 14 Sep 2026 09:28:00 +0000
Message-ID: <811965672782.299624@outlook.com>
MIME-Version: 1.0
Content-Type: text/plain; charset=utf-8
X-Forwarded-To: student@uni-example.edu
```